Highlights

The Beggining of Control Revolution: Ofset-free Koopman MPC

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The Beggining of Control Revolution: Ofset-free Koopman MPC

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Abstract

Keywords:

1. Introduction

2. Preliminaries and Notation

2.1. Offset-Free MPC

In this subsection, we will present a standard offset free model predictive control framework, introduced byTODO: cite pannocchia . This approach consist of state estimation - observer, target optimization and dynamic optimization. The whole approach is used to introduce integral action into model predictive control to help mitigate plant-model missmatch and ensure offset free control. If this compensation would not be used, the plant with this controller would not reach a desired reference (other than at origin), due to predictions using different model than actual plant.

The offset-free augmented system can be written in the following state-space form, including the disturbance input matrix B_d and disturbance output matrix C_d:

$$\begin{bmatrix} x_{k+1} \\ d_{k+1} \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} x_k \\ d_k \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k, \qquad y_k = \begin{bmatrix} C & C_d \end{bmatrix} \begin{bmatrix} x_k \\ d_k \end{bmatrix} \tag{1}$$

where x_k is the original system state, d_k is the estimated constant disturbance, u_k is the control input, y_k is the output, A, B, and C are the original state-space matrices, B_d and C_d encode how the disturbance enters the state and output equations, and I is the identity matrix. Here, n_x, n_u, n_y, and n_d denote the state, input, output, and disturbance dimensions, respectively, so that $x_k \in \mathbb{R}^{n_x}$, $u_k \in \mathbb{R}^{n_u}$, $y_k \in \mathbb{R}^{n_y}$, $d_k \in \mathbb{R}^{n_d}$, $A \in \mathbb{R}^{n_x \times n_x}$, $B \in \mathbb{R}^{n_x \times n_u}$, $C \in \mathbb{R}^{n_y \times n_x}$, $B_d \in \mathbb{R}^{n_x \times n_d}$, $C_d \in \mathbb{R}^{n_y \times n_d}$, and $I \in \mathbb{R}^{n_d \times n_d}$. For the offset free behavior, the number of the disturbances n_d must be lower or equal to the number of the measurements n_y. Also it is necessary to note that the number of offset free tracking is achievable only in the number of disturbances that are estimated. For the system where $n_d = n_y$ system and choice of the matrices B_d and C_d there is a necessary condition, that:

$$\operatorname{rank} \begin{bmatrix} I - A & -B_d \\ C & C_d \end{bmatrix} = n_x + n_y. \tag{2}$$

When solving the MPC, both states and disturbance are approximated by observer.

2.2. Koopman Operator

The koopman operator theory was introduced by TODO: cite and further popularize by TODO: cite . The theory enables to exactly model nonlinear systems linearly, by lifting its current states/measurements into infinite dimensional variables, where the governing operator for the evolving system is linear.

TODO: fancy smenši exuation

Over the years many approximations of the koopman operator from the data were proposed, starting with simple DMD, extended to the EDMD and recently the neural networks were used to best approximate the lifting functions so the system behaves linearly. The algorithms were as well developed not only for autonomous systems but for the systems with control as well TODO: cite and elaborate . Koopman operator in the MPC was introduced by TODO: cite Korda , and in recent years almost every kind of MPC was transfor to accept some kind of Koopman Operator, mostly represented as an LTI system in the state space form. The offset free wersion is no exception, in the work of TODO: Zotero, kukni papere, prosím ešte aj naskenuj ďalšie .

For the purpose of this work we are using the DeepKoopman Approximation introduced by TODO: cite, Lusch , transformed for the nonautonomous system. We use two kinds of approximations. Firstly the LTI approximation and secondly the approximation, where only the lifted system evolves linearly, but its projection is nonlinear (as it should be from the definition of Koopman operator).

Remark 1. For the

The architectures of those systems can be seen below. TODO: images

3. Koopman MPC with Offset-Free

In this subsection, we will present a standard offset free model predictive control framework using approximation of Koopman operator as a linear time invariant (LTI) model. As observer is standard Kalman filter, we will discuss mainly remaining two optimization problems. As we use the LTI approximation of Koopman operator, we can use the standard MPC formulation with linear constraints and objective function. The control structure is shown in Figure 1.

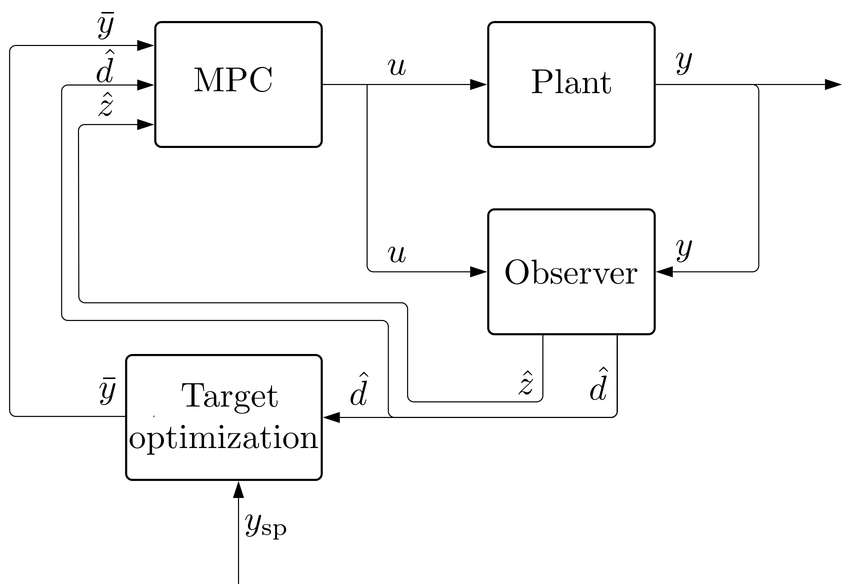

Figure 1: Basic closed-loop control structure.

The target optimization problem computes the steady-state targets $\bar{u}, \bar{z}, \bar{y}$ that minimizes the difference between reference y_{ref} and feasible output $\bar{y}$ in steady state. The problem is formulated as:

$$
\begin{aligned}
\min_{\bar{u},\bar{z},\bar{y}} \quad & (\bar{y} - y_{\text{ref}})^\intercal Q_{\text{y}} (\bar{y} - y_{\text{ref}}), && \text{(3a)}\\
\text{s.t.} \quad & \bar{z} = A\bar{z} + B\bar{u}, && \text{(3b)}\\
& \bar{y} = C\bar{z} + d, && \text{(3c)}\\
& u_{\min} \le \bar{u} \le u_{\max}, && \text{(3d)}\\
& y_{\min} \le \bar{y} \le y_{\max}, && \text{(3e)}\\
& d = \hat{d}(t), \quad y_{\text{ref}} = y_{\text{sp}}(t), && \text{(3f)}
\end{aligned}
$$

where $\bar{u} \in \mathbb{R}^{n_u}$, $\bar{z} \in \mathbb{R}^{n_z}$, and $\bar{y} \in \mathbb{R}^{n_y}$ are the steady-state input, lifted state, and output, respectively; $Q_{\text{y}} \succ 0$ is the output tracking weight matrix; $A \in \mathbb{R}^{n_z \times n_z}$, $B \in \mathbb{R}^{n_z \times n_u}$, $C \in \mathbb{R}^{n_y \times n_z}$ are the state-space matrices of the LTI Koopman model; $d = \hat{d}(t) \in \mathbb{R}^{n_y}$ is the additive output disturbance estimate provided by the Kalman filter; $y_{\text{ref}} = y_{\text{sp}}(t)$ is the output setpoint; and $u_{\min}, u_{\max}$, $y_{\min}, y_{\max}$ are the input and output constraint bounds.

Remark 2. For this

Given the steady-state target $(\bar{u}, \bar{z}, \bar{y})$, the MPC problem is solved at each sampling instant to compute the optimal input sequence over the horizon N:

$$
\begin{aligned}
\min_{u_0,\ldots,u_{N-1}} \quad & \sum_{k=0}^{N-1} (y_k - \bar{y})^\intercal Q_{\text{y}} (y_k - \bar{y}) + \sum_{k=0}^{N-1} \Delta u_k^\intercal Q_{\text{u}} \Delta u_k, && \text{(4a)}\\
\text{s.t.} \quad & z_{k+1} = A z_k + B u_k, \quad k \in \mathbb{N}_0^{N-1}, && \text{(4b)}\\
& y_k = C z_k + d, \quad k \in \mathbb{N}_0^{N-1}, && \text{(4c)}\\
& \Delta u_k = u_k - u_{k-1}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(4d)}\\
& u_{\min} \le u_k \le u_{\max}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(4e)}\\
& y_{\min} \le y_k \le y_{\max}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(4f)}\\
& z_0 = \hat{z}(t), \quad d = \hat{d}(t), \quad u_{-1} = u(t - T_{\text{s}}), && \text{(4g)}
\end{aligned}
$$

where $\hat{z}(t) \in \mathbb{R}^{n_z}$ is the lifted state estimate provided by the Kalman filter; and $u_{-1} = u(t - T_{\text{s}})$ is the input applied at the previous sampling instant, used to compute the first increment Δu_0. Following the receding-horizon principle, only the first element u_0^* of the optimal sequence is applied to the plant.

4. Methodology

Main drawback that we observe in a current Koopman MPC (offset-free, robust, or any other), is that the optimization is done without fully leveraging knowledge of the lifted space. Of course we are predicting in the lifted space, but in objective function we are using original space, that is created derived as linear transformation of the lifted space $y_k = Cz_k$. This transformation is not linear by nature $y_k = h(z_k)$, but to use linear MPC, we think we had no other choice. Some papers are not using this transformation directly, but after closer look, they are usually using equivalent transformations, for example expanding lifted space with measurements, but dynamic in lifted space is linear, so transformations to achieve back this measurements is linear. We will show that we can use the lifted space directly in the objective function, showing how to handle the lifted space and transformation of an objective function. We can fully eliminate the neccessity of the back transformation used in objective function in MPC, but we still need them to compute the target for the lifted space. In this case and case of constraints, we use a first order Taylor expansion, to better approximate the transformation.

4.1. Linearization of Output Mapping

For the control problem, we are not able to completely eliminate the neccessity of the transformation from the lifted space back to the original space. For this, we linearize the output mapping with first order taylor expansion. For the purpose of this work we experimented with linearization. at two different points: linearization at the previous target and linearization at the current state estimate.

$$y_k = h(z_k) \approx h(\bar{z}(t - T_\mathrm{s})) + J_{\bar{z}(t-T_\mathrm{s})}(z_k - \bar{z}(t - T_\mathrm{s}))\,, \tag{5a}$$

$$y_k = h(z_k) \approx h(\hat{z}(t)) + J_{\hat{z}(t)}(z_k - \hat{z}(t))\,, \tag{5b}$$

where $h : \mathbb{R}^{n_z} \to \mathbb{R}^{n_y}$ is the nonlinear output mapping; $\bar{z}(t-T_\mathrm{s})$ is the steady-state lifted state computed by the target optimizer at the previous sampling instant; $J_{\bar{z}(t-T_\mathrm{s})}$ and $J_{\hat{z}(t)}$ are the Jacobians of h evaluated at $\bar{z}(t - T_\mathrm{s})$ and $\hat{z}(t)$, respectively.

4.2. Target Optimization

The objective function of target estimation is the same as in the standard offset-free MPC. The main difference is in

The proposed target optimization problem is formulated as:

$$\begin{aligned}
\min_{\bar{u},\bar{z},\bar{y}} \quad & (\bar{y}-y_{\text{ref}})^\intercal Q_{\text{y}}(\bar{y}-y_{\text{ref}}), && \text{(6a)}\\
\text{s.t.} \quad & \bar{z} = A\bar{z} + B\bar{u}, && \text{(6b)}\\
& \bar{y} = H(z_{\text{LP}})\bar{z} + h(z_{\text{LP}}) - H(z_{\text{LP}})z_{\text{LP}} + d, && \text{(6c)}\\
& u_{\min} \leq \bar{u} \leq u_{\max}, && \text{(6d)}\\
& y_{\min} \leq \bar{y} \leq y_{\max}, && \text{(6e)}\\
& d = \hat{d}(t), \quad y_{\text{ref}} = y_{\text{sp}}, \quad z_{\text{LP}} = \bar{z}_{-1}, && \text{(6f)}
\end{aligned}$$

where $H(z_{\text{LP}})$ is the Jacobian of the transformation $h(z_{\text{LP}})$. z_{LP} is a linearization point, the point where we approximate our function with first order Taylor expansion. We propose, that this point should be as close as possible to the $\bar{z}$. Therefor we are using the previously calculated target $\bar{z}_{-1}$. This allow us to better approximate the transformation of the lifted space to the original space and more precise target optimization.

4.3. MPC

The proposed MPC problem is formulated as:

$$\begin{aligned}
\min_{u_0,\dots,u_{N-1}} \quad & \sum_{k=0}^{N-1}(z_k-\bar{z})^\intercal Q_{\text{z}}(z_k-\bar{z}) + \sum_{k=0}^{N-1}\Delta u_k^\intercal Q_{\text{u}}\Delta u_k, && \text{(7a)}\\
\text{s.t.} \quad & z_{k+1} = Az_k + Bu_k, \quad k \in \mathbb{N}_0^{N-1}, && \text{(7b)}\\
& y_k = H(z_{\text{LP}})z_k + h(z_{\text{LP}}) - H(z_{\text{LP}})z_{\text{LP}} + d, \quad k \in \mathbb{N}_0^{N-1}, && \text{(7c)}\\
& \Delta u_k = u_k - u_{k-1}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(7d)}\\
& u_{\min} \leq u_k \leq u_{\max}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(7e)}\\
& y_{\min} \leq y_k \leq y_{\max}, \quad k \in \mathbb{N}_0^{N-1}, && \text{(7f)}\\
& z_0 = \hat{z}(t), \quad d = \hat{d}(t), \quad u_{-1} = u(t-T_{\text{s}}) && \text{(7g)}
\end{aligned}$$

where $H(z_0)$ is the Jacobian of the transformation $h(z_0)$ at the point z_0. This linearization point is chosen to best approximate the surroundings of current operational point. The first order Taylor expansion is done to receive y_k, which is used only in the constrains of the system. In objective funtion, we use a tuning matric Q_{z} to penalize the difference from lifted space target. We propose to derive this tuning matrix as follows:

$$Q_{\text{z}} = H(\bar{z})^\intercal Q_{\text{y}} H(\bar{z}). \tag{8}$$

This allows us to use and tune our problem using matrix Q_{y}, which is convenient and easily tuned. This is a significant change from the standard Koopman MPC, where the objective function is defined in the original space.

4.4. Closed-Loop Control Structure

The close loop structure is similar to the previous with the few tweaks. There is an addition of the linearization block. Also there is a time varying Kalman filter as an observer and the optimization problems have a different formulation.

For the linearization at previous target the algorithm is:

Algorithm 1 Closed-loop control with nonlinear output mapping with linearization at previous target

---

Require: Koopman lifting function $g(\cdot)$, initial measurement $y(0)$

1: **Initialization:** $\hat{z}(0) \leftarrow g(y(0))$, $\hat{d}(0) \leftarrow 0$; solve (6) with $z_{\mathrm{LP}} = \hat{z}(0)$ and $\hat{d}(0)$ to obtain $\bar{z}_{-1}$
2: **loop**
3: **Measurement:** Obtain $y(t)$ from the plant
4: **Linearization:** Compute $H(\bar{z}_{-1})$ and $h(\bar{z}_{-1})$ — output mapping linearized at $z_{\mathrm{LP}} = \bar{z}_{-1}$
5: **Observer:** Run the time-varying Kalman filter with $y(t)$ and $u(t-T_{\mathrm{s}})$ to obtain updated estimates $\hat{z}(t)$ and $\hat{d}(t)$
6: **Target estimation:** Solve (6) with $\hat{d}(t)$ to obtain $\bar{z}(t)$, $\bar{u}(t)$, $\bar{y}(t)$; set $\bar{z}_{-1} \leftarrow \bar{z}(t)$
7: **MPC:** Solve (7) with $\hat{z}(t)$, $\hat{d}(t)$, $\bar{z}(t)$ to obtain the optimal first input u_0^*
8: **Apply input:** Apply $u(t) = u_0^*$ to the plant
9: **end loop**

---

where $g : \mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_z}$ is the Koopman lifting function that provides the initial lifted-state estimate, and $\hat{d}(0) = 0$ initialises the disturbance estimate at zero.

For the linearization at current estimate, it is:

Algorithm 2 Closed-loop control with nonlinear output mapping (linearization at current state estimate)

Require: Koopman lifting function $g(\cdot)$, initial measurement $y(0)$
1: **Initialization:** $\hat{z}(0) \leftarrow g(y(0))$, $\hat{d}(0) \leftarrow 0$
2: **loop**
3: **Measurement:** Obtain $y(t)$ from the plant
4: **Observer:** Run the time-varying Kalman filter with $y(t)$ and $u(t-T_\mathrm{s})$ to obtain updated estimates $\hat{z}(t)$ and $\hat{d}(t)$
5: **Linearization:** Compute $H(\hat{z}(t))$ and $h(\hat{z}(t))$ — output mapping linearized at $z_\mathrm{LP} = \hat{z}(t)$ TODO: when this is hapenning?
6: **Target estimation:** Solve (6) with $\hat{d}(t)$ to obtain $\bar{z}(t)$, $\bar{u}(t)$, $\bar{y}(t)$; set $\bar{z}_{-1} \leftarrow \bar{z}(t)$
7: **MPC:** Solve (7) with $\hat{z}(t)$, $\hat{d}(t)$, $\bar{z}(t)$ to obtain the optimal first input u_0^*
8: **Apply input:** Apply $u(t) = u_0^*$ to the plant
9: **end loop**

4.5. State Estimation - Time Varying Kalman Filter

The observer estimates the augmented state $\tilde{z}_k = [z_k^\intercal,\ d_k^\intercal]^\intercal$ from the noisy output y_k and the applied input u_k. The augmented linear dynamics are identical to (1) with x_k replaced by the lifted Koopman state z_k:

$$\tilde{z}_{k+1} = \tilde{A}\,\tilde{z}_k + \tilde{B}\,u_k, \qquad \tilde{A} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}, \tag{9}$$

where $\tilde{z}_k \in \mathbb{R}^{n_z+n_d}$ is the augmented state stacking the lifted Koopman state $z_k \in \mathbb{R}^{n_z}$ and the output disturbance estimate $d_k \in \mathbb{R}^{n_d}$, $\tilde{A} \in \mathbb{R}^{(n_z+n_d)\times(n_z+n_d)}$ and $\tilde{B} \in \mathbb{R}^{(n_z+n_d)\times n_u}$ are the augmented system matrices.

Because the output map is nonlinear, the measurement equation is relinearised at each sampling instant around a chosen linearisation point z_LP:

$$y_k \approx \tilde{C}_k\,\tilde{z}_k + c_k, \qquad \tilde{C}_k = \big[H(z_\mathrm{LP}),\ C_d\big], \quad c_k = h(z_\mathrm{LP}) - H(z_\mathrm{LP})\,z_\mathrm{LP}, \tag{10}$$

where $\tilde{C}_k \in \mathbb{R}^{n_y\times(n_z+n_d)}$ is the time-varying output matrix formed by stacking the Jacobian $H(z_\mathrm{LP}) \in \mathbb{R}^{n_y\times n_z}$ of h at the linearisation point and the disturbance output matrix C_d, and $c_k \in \mathbb{R}^{n_y}$ is the linearisation offset.

With the linear prediction model (9) and the time-varying output matrix (10), the filter runs the standard predict–update recursion at each step k:

Predict:

$$\hat{\tilde{z}}_{k|k-1} = \tilde{A}\,\hat{\tilde{z}}_{k-1|k-1} + \tilde{B}\,u_{k-1}, \tag{11a}$$

$$P_{k|k-1} = \tilde{A}\,P_{k-1|k-1}\,\tilde{A}^{\intercal} + Q, \tag{11b}$$

where $\hat{\tilde{z}}_{k|k-1} \in \mathbb{R}^{n_z+n_d}$ is the prior augmented-state estimate, $P_{k|k-1} \in \mathbb{R}^{(n_z+n_d)\times(n_z+n_d)}$ is the prior error covariance, and $Q \succeq 0$ is the process noise covariance matrix.

Update:

$$S_k = \tilde{C}_k\,P_{k|k-1}\,\tilde{C}_k^{\intercal} + R, \tag{12a}$$

$$K_k = P_{k|k-1}\,\tilde{C}_k^{\intercal}\,S_k^{-1}, \tag{12b}$$

$$\hat{\tilde{z}}_{k|k} = \hat{\tilde{z}}_{k|k-1} + K_k\Big(y_k - \tilde{C}_k\,\hat{\tilde{z}}_{k|k-1} - c_k\Big), \tag{12c}$$

$$P_{k|k} = \Big(I - K_k\,\tilde{C}_k\Big)\,P_{k|k-1}, \tag{12d}$$

where $S_k \in \mathbb{R}^{n_y\times n_y}$ is the innovation covariance, $K_k \in \mathbb{R}^{(n_z+n_d)\times n_y}$ is the Kalman gain, $\hat{\tilde{z}}_{k|k}$ is the posterior augmented-state estimate whose upper n_z entries give $\hat{z}(t)$ and lower n_d entries give $\hat{d}(t)$, $P_{k|k}$ is the posterior error covariance, and $R \succ 0$ is the measurement noise covariance matrix.

4.6. Block Diagonal Deep Koopman Transformation

TODO: bevare notation conflict zbar During our work, we encounter a several problems with the suboptimal or infeasible solutions using Koopman MPC using Gurobi solver. For the enhancement of the performaance, we are proposing a block diagonal structure of the matrix A. This transformation can be done after obtaining the Koopman operator, with finding the transformation matrix T that transforms the Koopman operator to the block diagonal form:

$$\bar{A} = T^{-1}AT. \tag{13}$$

This transformation is done by finding the eigenvalues and eigenvectors of the Koopman operator, and then grouping them into blocks. The transformation matrix T is constructed from the eigenvectors, and the block diagonal form $\bar{A}$ is obtained by applying the transformation to the original Koopman operator A. Using this approach, we had to modify the rest of the model accordingly, using the transformation matrix T to transform the input and output matrices B and C as well as the lifted states z:

$$\bar{B} = T^{-1}B, \quad \bar{C} = CT, \quad \bar{z}_k = T^{-1}z_k. \tag{14}$$

When computing the Jackobian, we are transforming the original Jackobian $H(z_k)$ to the block diagonal form $\bar{H}(z_k)$ using the transformation matrix T:

$$\bar{H}(z_k) = H(z_k)T = H(T\bar{z}_k)T. \tag{15}$$

For fair comparison, this transformation is applied to every identified model through this work, not only for Koopman models. The effects of this transformation are discussed in the result section of this work.

5. Identification

TODO: Specific parameters for each model that was identified and identification error comparison

6. Control Results

In the following, we make precise the output mappings sketched in the notes and list the combinations (setups) evaluated in the simulations.

6.1. The effect of block-diagonal transformation

After this transformation we are able to enhace the performance in optimization as can be seen in Table 1. This table is done using the formulation of Koopman MPC which can be seen in Sec. 3. The table shows the decrease in a computational times. Also with this formulation we are able to obtain optimal solution 100% time during simulation. Contrary, the full Koopman MPC formulation is able to obtain optimal solution only 26.7% of the time with the rest of the time giving warning, that the solution is suboptimal. We can see also increase in a control performance, which is measured by the objective function value, which is in optimal case arounf 6% lower. Also the settling time for both levels is lower.

Table 1: Comparison of structure of problem to solver time and control performance.

Structure	Time TE/MPC	Optimal? [%]	OBJ	ST h_1	ST h_2
Full	0.2392 / 2.3884	26.7	100.0	43	48
Block Diagonal	0.2191 / 1.6204	100.0	94.1	25	31

The second option how to obtain the block diagonal structure is to learn the Koopman operator directly in the block diagonal form. This can be done by limiting the number of learnable parameters in a Koopman matrix during training to those that are in the block diagonal form. This also speeds the training as less parameters are learned.

To obtain an optimal performance, which is neccessary for the real time control, we are proposing to use the block diagonal structure of the Koopman operator. There is no neccessity to use the full Koopman operator, as the block diagonal structure is able to capture the same dynamics of the system and favors the optimization performance. The way this structure is obtained, during or after the training, is not important, as long as the Koopman operator is in the block diagonal form.

6.2. Simulation setups

6.3. Simulation Results for Two Reactor System

We index target mappings by T1–T3 and dynamic mappings by D1–D4 as in (**??**)–(**??**). Table 2 compares previous diagonal entries (CT, T2D2, T3D3) with reruns from the current scripts.

Table 2: Two-reactor CSTR: normalized closed-loop objective with NMPC benchmark 100. Diagonal combinations CT ($T1 \times D1$), T2D2 ($T2 \times D2$), and T3D3 ($T3 \times D3$). *Previous*: entries from the earlier full 3×4 TM/DM matrices. *Updated*: $100\,J/J_{\mathrm{NMPC}}$ from `control_cstr` (`CT.py`, `T2D2.py`, `T3D3.py`) with the same weighting as tuning I; em dashes mark tunings II–III not yet rerun.

	Previous			**Updated**		
Tuning	CT	T2D2	T3D3	CT	T2D2	T3D3
I: $Q_y = \mathrm{diag}(5)$, $Q_u = \mathrm{diag}(0.1)$	365.0	251.5	265.1	314.9	260.5	285.7
II: $Q_y = Q_u = I$	184.5	146.2	153.9	—	—	—
III: $Q_y = \mathrm{diag}(0.1)$, $Q_u = \mathrm{diag}(5)$	130.4	112.1	116.1	—	—	—

7. Discussion

8. Conclusion

Appendix A. System Models

In this appendix, we provide detailed mathematical descriptions of the two benchmark nonlinear systems used to evaluate the proposed offset-free Koopman MPC schemes: a two-tank system with hydraulic flows and a two-reactor CSTR series with recycle and parallel reactions.

Appendix A.1. Two Reactor System

The two-reactor system consists of two continuous stirred tank reactors (CSTRs) in series with a recycle loop from the second reactor back to the first. Each reactor is equipped with jacket cooling. The system models a classic selectivity problem with parallel competing reactions on reactant A.

Appendix A.1.1. Chemical Reactions

Three reaction pathways are considered:

$$
\begin{aligned}
\text{(R1) Desired:} \quad & 2\,\mathrm{A} \rightarrow \mathrm{B}, && r_1 = k_1(T)\, C_A^2, && \text{(A.1)}\\
\text{(R2) Undesired:} \quad & \mathrm{A} \rightarrow \mathrm{U}, && r_2 = k_2(T)\, C_A, && \text{(A.2)}\\
\text{(R3) Shunt (R1 only):} \quad & \mathrm{A} + \mathrm{B} \rightarrow 2\,\mathrm{B}, && r_3 = k_3(T)\, C_A\, C_B, && \text{(A.3)}
\end{aligned}
$$

where the temperature-dependent rate constants follow Arrhenius kinetics:

$$k_i(T) = k_{i,0}\, \exp\left(-\frac{E_i}{RT}\right), \quad i = 1, 2, 3. \tag{A.4}$$

The autocatalytic shunt reaction (R3) occurs only in the first reactor.

Appendix A.1.2. Dynamic Model

The system has eight states:

$$x = \begin{bmatrix} C_{A,1} & T_1 & C_{A,2} & T_2 & C_{B,1} & C_{B,2} & C_{U,1} & C_{U,2} \end{bmatrix}^\top, \tag{A.5}$$

where $C_{A,i}$, $C_{B,i}$, and $C_{U,i}$ denote concentrations of species A, B, and U in reactor i, and T_i is the temperature. The four control inputs are:

$$u = \begin{bmatrix} F & L & T_{c,1} & T_{c,2} \end{bmatrix}^\top, \tag{A.6}$$

with F the fresh feed flow rate, L the recycle flow rate, and $T_{c,i}$ the jacket coolant temperatures.

The continuous-time dynamics are:

$$\frac{dC_{A,1}}{dt} = \frac{1}{V_1}\big[C_{A,0}\,F + L\,C_{A,2} - (F+L)\,C_{A,1}\big] - 2\,r_{1,1} - r_{2,1} - r_{3,1}, \tag{A.7a}$$

$$\frac{dC_{B,1}}{dt} = \frac{1}{V_1}\big[C_{B,0}\,F + L\,C_{B,2} - (F+L)\,C_{B,1}\big] + r_{1,1} + r_{3,1}, \tag{A.7b}$$

$$\frac{dC_{U,1}}{dt} = \frac{1}{V_1}\big[C_{U,0}\,F + L\,C_{U,2} - (F+L)\,C_{U,1}\big] + r_{2,1}, \tag{A.7c}$$

$$\begin{aligned}\frac{dT_1}{dt} &= \frac{1}{V_1}\big[T_0\,F + L\,T_2 - (F+L)\,T_1\big] - \frac{U_1 A_1}{V_1\,\rho\,c_p}(T_1 - T_{c,1}) \\ &\quad + \frac{1}{\rho\,c_p}\big[(-\Delta H_1)\,r_{1,1} + (-\Delta H_2)\,r_{2,1} + (-\Delta H_3)\,r_{3,1}\big],\end{aligned} \tag{A.7d}$$

$$\frac{dC_{A,2}}{dt} = \frac{F+L}{V_2}\big[C_{A,1} - C_{A,2}\big] - 2\,r_{1,2} - r_{2,2}, \tag{A.7e}$$

$$\frac{dC_{B,2}}{dt} = \frac{F+L}{V_2}\big[C_{B,1} - C_{B,2}\big] + r_{1,2}, \tag{A.7f}$$

$$\frac{dC_{U,2}}{dt} = \frac{F+L}{V_2}\big[C_{U,1} - C_{U,2}\big] + r_{2,2}, \tag{A.7g}$$

$$\begin{aligned}\frac{dT_2}{dt} &= \frac{F+L}{V_2}\big[T_1 - T_2\big] - \frac{U_2 A_2}{V_2\,\rho\,c_p}(T_2 - T_{c,2}) \\ &\quad + \frac{1}{\rho\,c_p}\big[(-\Delta H_1)\,r_{1,2} + (-\Delta H_2)\,r_{2,2}\big],\end{aligned} \tag{A.7h}$$

where $r_{j,i}$ denotes the rate of reaction j in reactor i. The outputs are all eight states:

$$y = x. \tag{A.8}$$

The system is discretized with sampling time $T_s = 1\,\mathrm{s}$ using a Runge–Kutta integration scheme. Physical parameters, kinetic constants, and operational constraints are listed in Tables A.3–A.5.

Table A.3: CSTR series physical and thermodynamic parameters.

Parameter	**Description**	**Value**
V_1	Volume of reactor 1	$1.0 \times 10^{-3}\,\mathrm{m}^3$
V_2	Volume of reactor 2	$2.0 \times 10^{-3}\,\mathrm{m}^3$
$U_1 A_1$	Heat transfer coefficient × area (R1)	$0.461\,\mathrm{kJ/(s{\cdot}K)}$
$U_2 A_2$	Heat transfer coefficient × area (R2)	$0.732\,\mathrm{kJ/(s{\cdot}K)}$
ρ	Liquid density	$1050\,\mathrm{kg/m}^3$
c_p	Specific heat capacity	$3.766\,\mathrm{kJ/(kg{\cdot}K)}$
R	Universal gas constant	$8.3145 \times 10^{-3}\,\mathrm{kJ/(mol{\cdot}K)}$
$C_{A,0}$	Feed concentration of A	$97.35\,\mathrm{mol/m}^3$
$C_{B,0}, C_{U,0}$	Feed concentrations of B, U	$0.0\,\mathrm{mol/m}^3$
T_0	Feed temperature	$298.0\,\mathrm{K}$

Table A.4: Arrhenius kinetic parameters for reactions R1–R3.

Reaction	**Parameter**	**Pre-exponential**	**Activation Energy**	**Heat of Reac**
R1: $2\,\mathrm{A} \rightarrow \mathrm{B}$	$k_{1,0},\ E_1,\ \Delta H_1$	$1.0 \times 10^5\,\mathrm{m}^3/(\mathrm{mol{\cdot}s})$	$45.0\,\mathrm{kJ/mol}$	60.0 kJ/mo
R2: $\mathrm{A} \rightarrow \mathrm{U}$	$k_{2,0},\ E_2,\ \Delta H_2$	$9.8 \times 10^9\,\mathrm{s}^{-1}$	$70.0\,\mathrm{kJ/mol}$	40.0 kJ/mo
R3: $\mathrm{A} + \mathrm{B} \rightarrow 2\,\mathrm{B}$	$k_{3,0},\ E_3,\ \Delta H_3$	$5.0 \times 10^4\,\mathrm{m}^3/(\mathrm{mol{\cdot}s})$	$55.0\,\mathrm{kJ/mol}$	60.0 kJ/mo

Table A.5: CSTR series operational constraints and initial conditions.

Parameter	**Description**	**Value/Range**
$F \in$	Fresh feed flow rate	$[0.0, 2.0 \times 10^{-4}]\,\mathrm{m}^3/\mathrm{s}$
$L \in$	Recycle flow rate	$[0.0, 2.0 \times 10^{-4}]\,\mathrm{m}^3/\mathrm{s}$
$T_{c,1}, T_{c,2} \in$	Coolant temperatures	$[280.0, 330.0]\,\mathrm{K}$
x_0	Initial state (steady state)	Computed for $u_0 = [1.5 \times 10^{-4}, 7.5 \times 10^{-5}, 290, 290]^\top$
T_s	Sampling time	$1.0\,\mathrm{s}$